Module 3 Homework 2 Challenge

Provide the best answer for each of the following questions.

Time Remaining 00:59:13

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

1

Review Later

Jenny has taken 5 tests for her geometry class and has an average score of 83. If she wants an average score of 85, what does she need to get on her next test?

2

Review Later

If number a decreases by 20% and number b increases by 10%,
then which of the following expressions is equal to the sum of a and b
after they have changed?

- $a - 20\% + b + 10\%$
- $0.20a + 0.10b$
- $(a + b) \times (-10\%)$
- $0.8a + 1.1b$
- $1.2a + 1.1b$

3

Review Later

The average of Andrew, Beatrice, and Carl's ages is 29. What will the average of their ages be in exactly 4 years?

4

Review Later

If an arithmetic sequence has a as its first term, which of the following is the nth term of this sequence? (Here r is the difference between neighboring terms.)

- nth term $= a \times n$
- nth term $= a^n$
- nth term $= a \times r^n$
- nth term $= a + (n - 1)r$
- nth term $= a + n$

5

Review Later

Find the value of $\sqrt{(\sqrt{7})^3}$ to the power 4.

6

Review Later

The simplified version of $\sqrt{12+6\sqrt{3}}$ should have which of the following forms, where a and b are both integers, and the square root is in simplest form?

- $a\sqrt{3}+b\sqrt{2}$
- $a\sqrt{3}$
- $a+b\sqrt{3}$
- a
- $a\sqrt{b}$

7

Review Later

Find the fraction that is the same as $0.133133133...$ Your answer should be a fraction, without spaces, e.g. 1/2

8

Review Later

When a decimal point is placed between two consecutive numbers, the new number that is formed is their average. What is the sum of the two numbers?

9

Review Later

Simplify this expression: $\sqrt{12+6\sqrt{3}}$

- $3+\sqrt{3}$
- $4\sqrt{3}$
- $3\sqrt{3}$
- $6+3\sqrt{3}$
- $3+6\sqrt{3}$

10

Review Later

Given two numbers a and b, where $a \neq b$, what is true about the geometric mean compared to the arithmetic mean of a and b?

- The geometric mean is always larger than the arithmetic mean
- The geometric mean is always smaller than the arithmetic mean
- The geometric mean is always the same as the arithmetic mean
- The geometric mean is smaller if a and b are less than 10

- [ ] The geometric mean is larger if a and b are greater than 10

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11

Review Later

The third and twenty-seventh terms of an arithmetic sequence are -51 and 21, respectively. What is the seventy-second term?

12

Review Later

If one arithmetic sequence has first term a and difference between terms r, and a second arithmetic sequence has first term $a+1$ and difference between terms equal to $r+1$, then find the difference between the sum of the first 2019 terms of the first sequence and the first 2019 terms of the second sequence.

- 1
- 2
- $2019+1+2+3+4+\ldots+2017+2018$
- 2019
- $1+2+3+4+\ldots+2017+2018$

13

Review Later

If $(\frac{1}{4})^4 = 2^x$, find the value of x.

14

Review Later

Which is bigger, $\left(\left((2^2)^2\right)^2\right)^2$ or $2^{2^{2^{2^2}}}$?

- $\left(\left((2^2)^2\right)^2\right)^2$
- $2^{2^{2^{2^2}}}$
- They are the same
- Not enough information to tell

15

Review Later

Find the fractional form of the decimal $0.2\overline{777}$.

Your answer should be a fraction in simplest terms without spaces, e.g. 1/2

16

Review Later

Find the value of

$$1.142857142857\ldots+2.285714285714\ldots+3.428571428571\ldots+4.571428571428\ldots+5.714285714285\ldots+6.857142857142\ldots+$$
$$7.142857142857\ldots+8.285714285714\ldots+9.428571428571\ldots+10.571428571428\ldots+11.714285714285\ldots+12.857142857142\ldots+$$
$$13.142857142857\ldots+14.285714285714\ldots+15.428571428571\ldots+16.571428571428\ldots+17.714285714285\ldots+18.857142857142\ldots+$$
$$19.142857142857\ldots+20.285714285714\ldots+21.428571428571\ldots+22.571428571428\ldots+23.714285714285\ldots+24.857142857142\ldots$$

17

Review Later

Find the value of $(\sqrt[3]{2}+\sqrt[3]{16}+\sqrt[3]{54})^3$.

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18

Review Later

Simplify this expression: $\sqrt{\sqrt{5}-\sqrt{3-\sqrt{29-12\sqrt{5}}}}$

19

Review Later

The sum of the first 4 terms in an arithmetic series is 58. The sum of the first 8 terms in the arithmetic series is 236. What is the first term of the series? Your answer should be a fraction without spaces, e.g. 1/2

20

Review Later

For each of the integers from 1 to 2019, take the greatest positive integer less than or equal to the square root of it. Find the sum of all these numbers. Write the number without commas, e.g. 11999

View Marked Questions

Finished

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